A terminal-tower reduction for high Collatz decoder depth

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Let

$$2^{a_i} x_{i+1} = 3x_i + 1, \qquad a_i = v_2(3x_i + 1), \tag{1}$$

where the x_i are positive odd integers. Put

$$A_k := \sum_{i=0}^{k-1} a_i, \qquad b_i := v_3(x_i + 1), \qquad x_i + 1 = 3^{b_i} u_i, \qquad 3 \nmid u_i. \tag{2}$$

We call the exponent block $(a_0, \dots, a_{r-1})$ a strict barrier block if

$$2^{A_k} < 3^k \qquad (1 \le k \le r). \tag{3}$$

Set $\alpha := \log_2 3$. For $a \ge 1$, define

$$q(a) := v_3(2^a - 2), \qquad q(1) := \infty. \tag{4}$$

Lemma 1 (one-step source dichotomy)**.** *Fix one step, and write $a := a_i$, $b := b_i$, and $b' := b_{i+1}$. If $q(a) \ne b + 1$, then*

$$b' = \min\{q(a), b + 1\}. \tag{5}$$

If $q(a) = b + 1 < \infty$, put

$$\rho_a := \frac{2^a - 2}{3^{q(a)}}. \tag{6}$$

Then

$$b' = b + 1 + v_3(u_i + \rho_a). \tag{7}$$

Moreover,

$$q(a) = \begin{cases} 0, & a \text{ is even}, \\ 1 + v_3((a-1)/2), & a > 1 \text{ is odd}. \end{cases} \tag{8}$$

Proof. Adding $2^a - 1$ to (1) gives

$$2^a(x_{i+1} + 1) = 3(x_i + 1) + 2^a - 2. \tag{9}$$

The two terms on the right have valuations $b+1$ and $q(a)$. Distinct valuations give (5), while equal finite valuations give (7) after division by 3^{b+1}.

If a is even, then $2^a - 2 \not\equiv 0 \pmod 3$. If $a = 2n + 1 > 1$, then

$$2^a - 2 = 2(4^n - 1).$$

Writing $n = 3^s m$ with $3 \nmid m$, the congruence $(1 + 3)^m - 1 \equiv 3m \pmod 9$ gives $v_3(4^m - 1) = 1$. The factorization $X^3 - 1 = (X - 1)(X^2 + X + 1)$, with $X^2 + X + 1 \equiv 3 \pmod 9$ for $X \equiv 1 \pmod 3$, raises the valuation by one whenever the exponent is multiplied by 3. This proves (8). □

Theorem 2 (terminal tower or terminal collision)**.** *Let $(a_0, \ldots, a_{r-1})$ be a strict barrier block, and define*

$$K(r) := \left\lceil \log_3 \left(\frac{\alpha r}{2} \right) \right\rceil. \tag{10}$$

Let ℓ be the length of its terminal run of exponent-one letters. If $\ell = r$, then

$$b_r = b_0 + r. \tag{11}$$

If $\ell < r$, put $s := r - \ell - 1$, so $a_s > 1$, and define

$$g_s := \begin{cases} v_3(u_s + \rho_{a_s}), & q(a_s) = b_s + 1, \\ 0, & q(a_s) \neq b_s + 1. \end{cases} \tag{12}$$

Then

$$\boxed{b_r \leq K(r) + \ell + g_s.} \tag{13}$$

Thus every decoder depth above the logarithmic scale $K(r)$ comes from the terminal exponent-one tower, from the final non-one collision, or from both.

Proof. Condition (3) gives

$$a_i \leq A_{i+1} < \alpha(i + 1) \leq \alpha r. \tag{14}$$

For every odd $a_i > 1$, equation (8) gives

$$3^{q(a_i)-1} \leq \frac{a_i - 1}{2} < \frac{\alpha r}{2} \leq 3^{K(r)}.$$

Hence $q(a_i) \le K(r)$. The same bound holds for even a_i, since then $q(a_i) = 0$.

Suppose $\ell < r$. If the step at s is not a collision, (5) gives

$$b_{s+1} \le q(a_s) \le K(r). \tag{15}$$

If it is a collision, (7) and $q(a_s) = b_s + 1$ give

$$b_{s+1} = q(a_s) + g_s \le K(r) + g_s. \tag{16}$$

Every one of the remaining ℓ exponent-one steps raises the decoder depth by exactly one. Equations (15) and (16) therefore prove (13). The same one-step law proves (11). □

Corollary 3 (deficit-three rate reduction)**.** *Under the hypotheses of Theorem 2, suppose that the terminal decoder has deficit at most three:*

$$b_r \ge r - 3. \tag{17}$$

If the block is not all ones, then

$$\boxed{\ell + g_s \ge r - 3 - K(r).} \tag{18}$$

In particular, for every $0 \le \eta < 1$*,*

$$\ell \le \eta r \quad \Longrightarrow \quad g_s \ge (1 - \eta)r - 3 - K(r). \tag{19}$$

Whenever the right side of (19) is positive, the last non-one step obeys the explicit collision conditions

$$q(a_s) = b_s + 1, \qquad u_s \equiv -\rho_{a_s} \pmod{3^{g_s}}. \tag{20}$$

Consequently, along any family of strict barrier blocks satisfying (17), a sublinear terminal exponent-one tower forces a collision of depth $r - o(r)$ *at the last non-one step.*

Proof. Equation (18) is (13) rearranged under (17), and (19) follows immediately. If its right side is positive, then $g_s > 0$. Definition (12) gives both claims in (20). Since $K(r) = O(\log r)$, the final assertion follows from (19). □